SBE37-SMP MicroCAT

Instrument Configuration

Instrument Serial Number:	37-16241
Instrument Firmware Version:	4.1
Zero Conductivity Frequency:	2758.15
Communications Format:	RS232
Communications Settings:	9600 baud, 8 Data Bits, No Parity

Installed Devices/Sensors

Data Format	*Measurement*	*Sensor Type*	*Serial Number*	*Rating*
Count	Temperature	Internal	N/A	N/A
Frequency	Conductivity	Internal	N/A	N/A
Count	Pressure Sensor	Druck	10746416	2000m(2000 dBar)

Maximum Depth: **2000m**

CAUTION - The maximum deployment depth will be limited by the measurement range of the pressure sensor, if installed, an attached sensor, if installed, or the housing.

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Bellevue, WA 98005
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seabird@seabird.com
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SENSOR SERIAL NUMBER: 16241
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 TEMPERATURE CALIBRATION DATA
ITS-90 TEMPERATURE SCALE

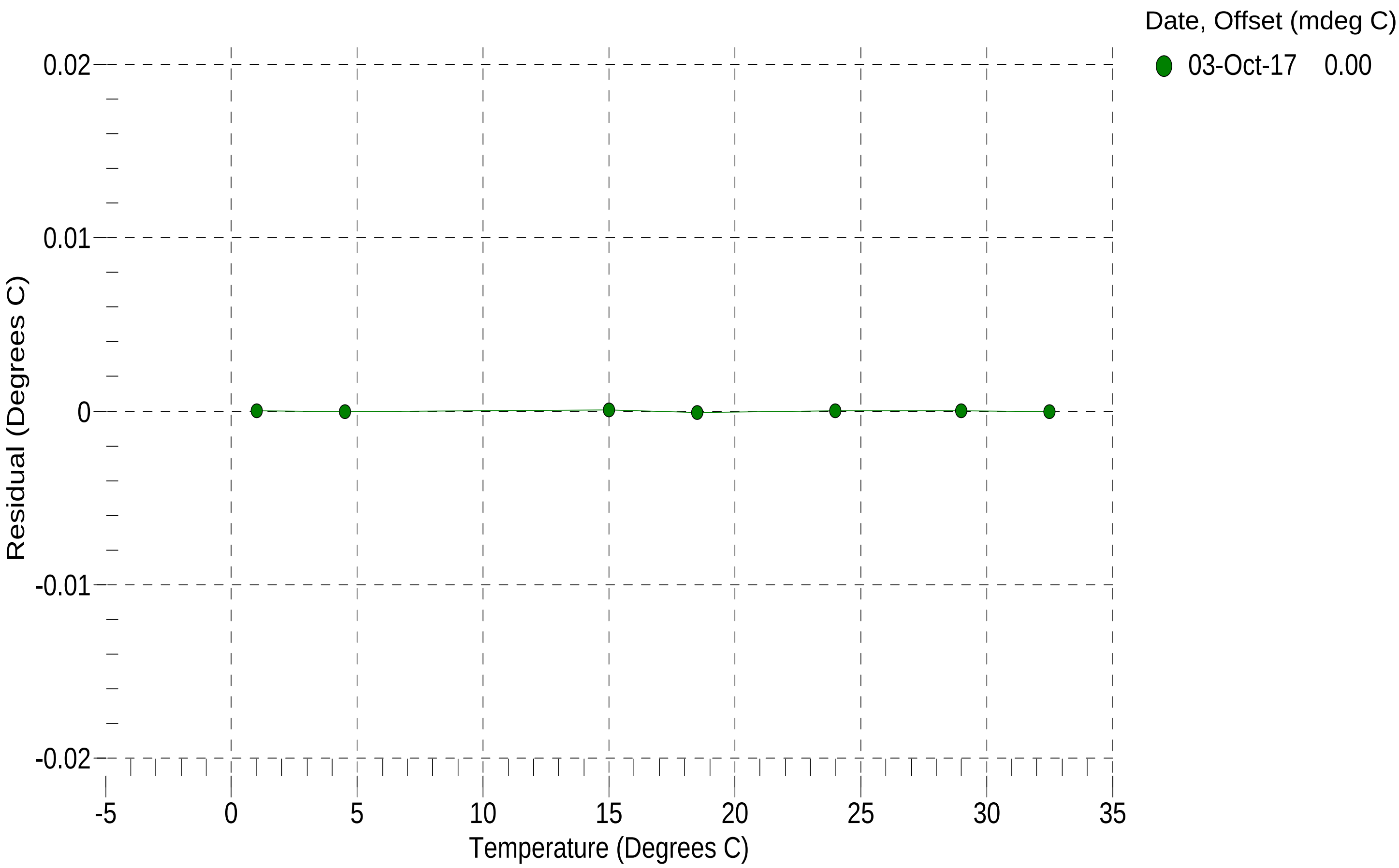
COEFFICIENTS:
a0 = -2.219656e-004
a1 = 3.186392e-004
a2 = -4.810768e-006
a3 = 2.127656e-007

BATH TEMP (° C)	INSTRUMENT OUTPUT (counts)	INST TEMP (° C)	RESIDUAL (° C)
1.0000	563013.1	1.0000	0.0000
4.5000	483644.8	4.5000	-0.0000
15.0000	312522.2	15.0001	0.0001
18.5000	271841.9	18.4999	-0.0001
24.0000	219621.9	24.0000	0.0000
29.0000	181990.9	29.0000	0.0000
32.5000	160075.9	32.5000	-0.0000

n = Instrument Output (counts)

Temperature ITS-90 (°C) = $1/\{a0 + a1[ln(n)] + a2[ln^2(n)] + a3[ln^3(n)]\} - 273.15$

Residual (°C) = instrument temperature - bath temperature



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SENSOR SERIAL NUMBER: 16241
CALIBRATION DATE: 03-Oct-17

SBE 37 V2 CONDUCTIVITY CALIBRATION DATA
PSS 1978: C(35,15,0) = 4.2914 Siemens/meter

COEFFICIENTS:

```
g =  -9.795033e-001
h =   1.288273e-001
i =  -8.863383e-005
j =   2.285287e-005

CPcor = -9.5700e-008
CTcor =  3.2500e-006
WBOTC =  9.0633e-008
```

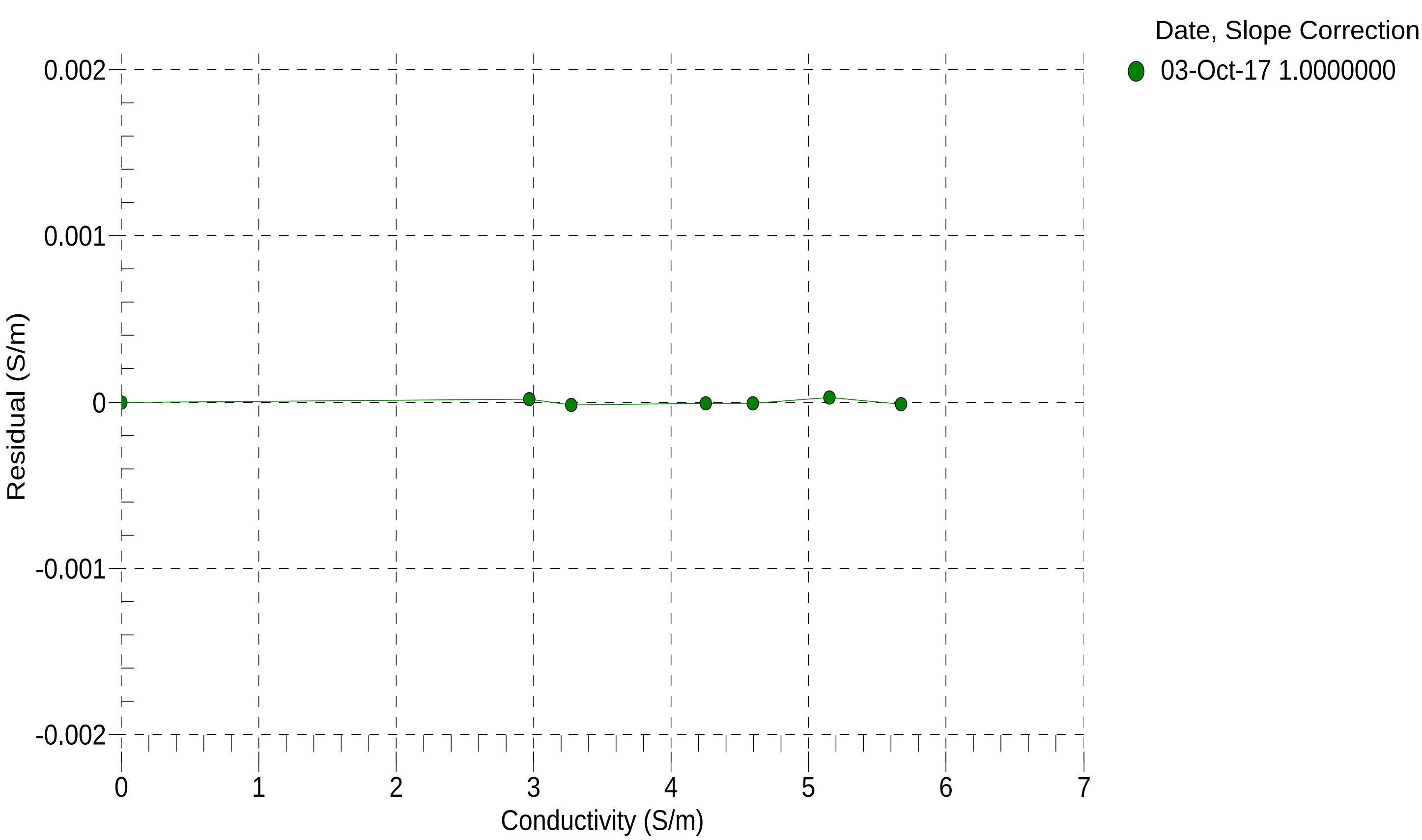
BATH TEMP (° C)	BATH SAL (PSU)	BATH COND (S/m)	INSTRUMENT OUTPUT (Hz)	INSTRUMENT COND (S/m)	RESIDUAL (S/m)
22.0000	0.0000	0.00000	2758.15	0.00000	0.00000
1.0000	34.6932	2.96644	5529.95	2.96646	0.00002
4.5000	34.6732	3.27255	5739.66	3.27253	-0.00002
15.0000	34.6306	4.25124	6363.23	4.25123	-0.00001
18.5000	34.6218	4.59534	6568.15	4.59534	-0.00001
24.0000	34.6123	5.15162	6886.28	5.15165	0.00003
29.0000	34.6073	5.67193	7170.72	5.67191	-0.00001
32.5000	34.6050	6.04330	7366.78	6.04313	-0.00017

f = Instrument Output(Hz) * sqrt(1.0 + WBOTC * t) / 1000.0

t = temperature (°C);   p = pressure (decibars); δ = CTcor; ε = CPcor;

Conductivity (S/m) = $(g + h * f^2 + i * f^3 + j * f^4) / (1 + \delta * t + \varepsilon * p)$

Residual (Siemens/meter) = instrument conductivity - bath conductivity



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SENSOR SERIAL NUMBER: 16241
CALIBRATION DATE: 27-Sep-17

SBE 37 V2 PRESSURE CALIBRATION DATA
2900 psia S/N 10746416

COEFFICIENTS:

PA0 =	4.085258e-001	PTCA0 =	5.245705e+005
PA1 =	8.923309e-003	PTCA1 =	2.222657e+000
PA2 =	-1.544797e-010	PTCA2 =	-1.530455e-002
PTEMPA0 =	-7.633685e+001	PTCB0 =	2.523101e+001
PTEMPA1 =	4.802460e-002	PTCB1 =	-9.975062e-004
PTEMPA2 =	-1.936485e-007	PTCB2 =	0.000000e+000

PRESSURE SPAN CALIBRATION

PRESSURE (PSIA)	INSTRUMENT OUTPUT (counts)	THERMISTOR OUTPUT (volts)	COMPUTED PRESSURE (PSIA)	RESIDUAL (%FSR)
14.63	526213.5	2086.2	14.70	0.00
591.69	590877.5	2088.4	591.56	-0.00
1168.83	655729.3	2089.6	1168.81	-0.00
1746.03	720728.2	2090.4	1746.05	0.00
2323.18	785863.4	2091.1	2323.20	0.00
2900.19	851116.9	2091.7	2900.08	-0.00
2323.21	785879.6	2091.2	2323.34	0.00
1746.04	720730.8	2090.8	1746.08	0.00
1168.98	655749.2	2090.3	1168.98	0.00
591.65	590878.0	2089.9	591.57	-0.00
14.62	526212.3	2092.0	14.68	0.00

THERMAL CORRECTION

TEMP (°C)	THERMISTOR OUTPUT (volts)	INSTRUMENT OUTPUT (counts)
32.50	2287	526309.16
29.00	2213	526305.53
24.00	2107	526299.75
18.50	1991	526289.42
15.00	1917	526281.51
4.50	1695	526265.60
1.00	1621	526254.69

TEMPERATURE (°C)	SPAN (mV)
-5.00	25.24
35.10	25.20

y = thermistor output (counts)

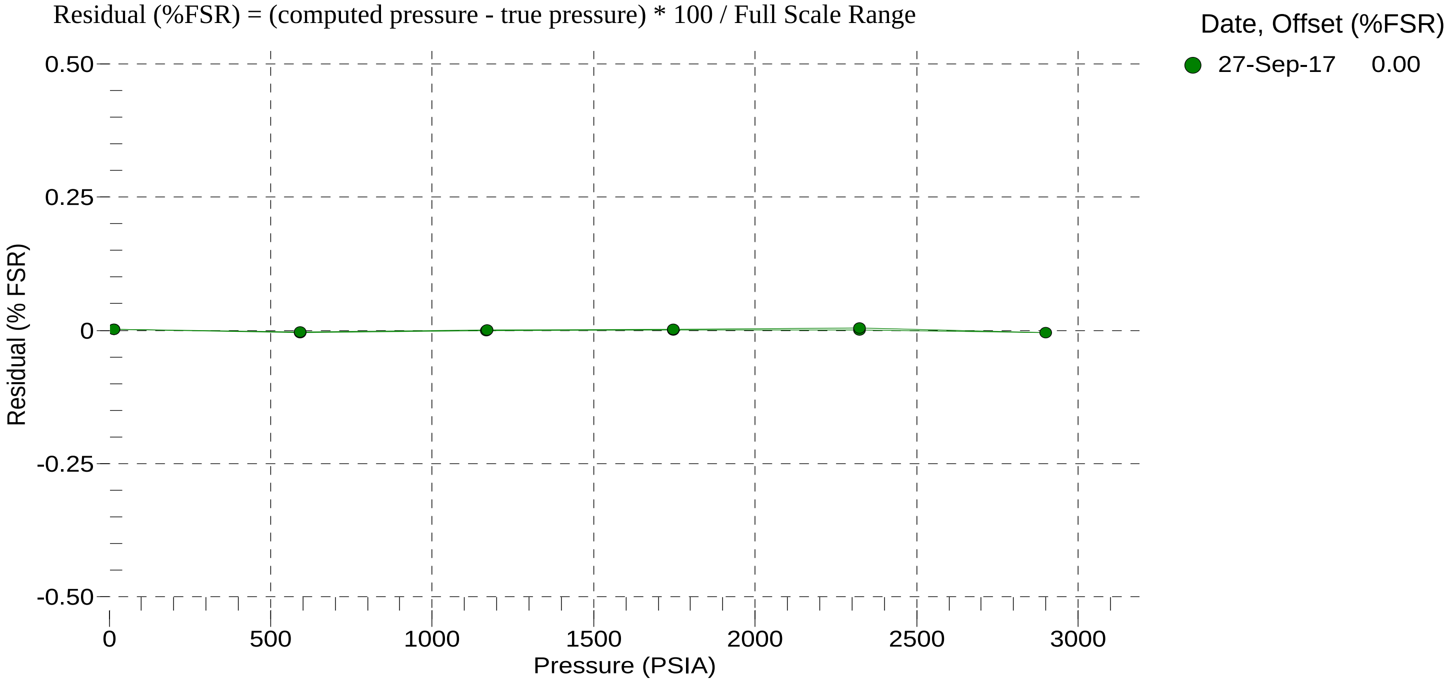
$t = PTEMPA0 + PTEMPA1 * y + PTEMPA2 * y^2$

$x = \text{instrument output} - PTCA0 - PTCA1 * t - PTCA2 * t^2$

$n = x * PTCB0 / (PTCB0 + PTCB1 * t + PTCB2 * t^2)$

$\text{pressure (PSIA)} = PA0 + PA1 * n + PA2 * n^2$

Residual (%FSR) = (computed pressure - true pressure) * 100 / Full Scale Range



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Pressure Test Certificate

Test Date: 2017-10-09 Description: SBE-37 Microcat

Sensor Information:

Model Number: SBE-37

Serial Number: 16241

Pressure Test Protocol:

Low Pressure Test: 40	PSI	Held For:	15	Minutes
High Pressure Test: 2900	PSI	Held For:	15	Minutes

Passed Test: True

Tested By: TH